

Mechanical Power Module Compliance Authority Framework

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Associated Baseline Documents: MPM-CHR-2026-001 (Charter), MPM-GOV-2026-002 (Governance Structure), MPM-RMP-2026-003 (Standardization Roadmap)

Executive Summary

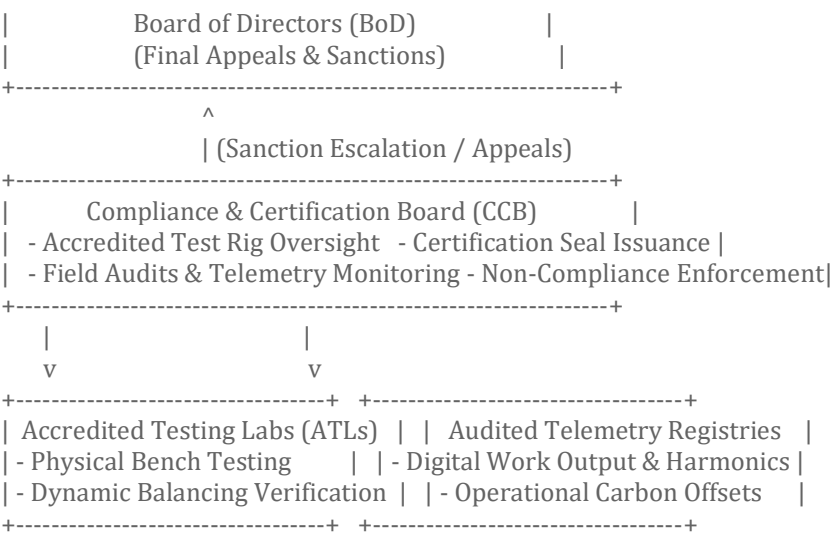
The **MPM Compliance Authority Framework (CAF)** defines the structural mandate, audit protocols, laboratory test procedures, and legal mechanisms used to certify physical hardware and software telemetry within the Mechanical Power Module Consortium ecosystem.

Administered directly by the **Compliance & Certification Board (CCB)**, this framework ensures that all physical 77 kW modules, 1 MW prime movers, containerized Multi-MPM Aggregator Adapters ($10\text{ MW} / 7.2\text{ GWh/month}$ blocks), and dedicated off-grid industrial interfaces strictly conform to ratified Consortium standards (RCS). Through third-party accredited testing and continuous operational telemetry, the CAF protects the commercial integrity of the **Consortium Interoperability Seal**.

1. Compliance Architecture & Operational Mandate

The CCB operates as an independent regulatory body under the authority of the Board of Directors (BoD) and technical guidance of the Technical Steering Committee (TSC).

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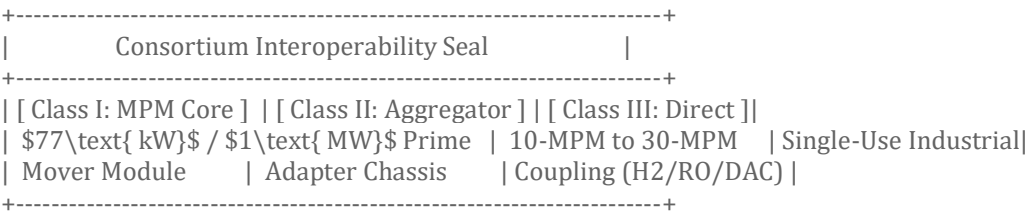


1.1 Scope of Authority

- 1. **Accreditation of Testing Facilities:** Qualification and licensing of independent Accredited Testing Laboratories (ATLs) equipped with standardized multi-input mechanical test rigs.
- 2. **Hardware Certification:** Testing and approval of baseline MPM modules, aggregator gearboxes, thermal management loops, and single-use industrial couplings.
- 3. **Seal Administration:** Granting, monitoring, and revoking the **Consortium Interoperability Seal**.
- 4. **Market Surveillance & Field Audits:** Random field inspection, telemetry data validation, and teardown audits of commercial production units.

2. Certification Classes & Seal Badging

To accommodate diverse manufacturing capabilities and deployment modes, the CCB issues three distinct certification classes:



2.1 Class I: Prime Mover Module Certification

Applies to individual \$77\text{ kW}\$ baseline units and \$1\text{ MW}\$ upscaled single-unit prime movers.

- **Requirements:** Strict compliance with spline geometry (MPM-SPEC-101), container lock points (MPM-SPEC-102), dynamic balance limits (MPM-SPEC-103), and auxiliary cooling quick-disconnects (MPM-SPEC-203).

2.2 Class II: Multi-MPM Aggregator Adapter Certification

Applies to containerized Multi-MPM Adapter chassis, multi-input gear trains, and 3-high stacking frameworks.

- **Requirements:** Verification of multi-shaft intake dynamics (MPM-SPEC-201), torsional vibration isolation (MPM-SPEC-202), 1 MW thermal rejection capacity (MPM-SPEC-203), and 3-high physical load ratings (MPM-SPEC-401).

2.3 Class III: Dedicated Single-Use Integration Certification

Applies to specialized direct-coupling machinery operating off-grid.

- **Requirements:** Validation of mechanical-to-fluid interfaces for megawatt electrolyzers (MPM-SPEC-301), high-pressure RO desalination pumps (MPM-SPEC-302), and high-volume Direct Air Capture fan drives (MPM-SPEC-303).

3. Physical Test Protocols & Compliance Metrics

Candidates submitting hardware for certification must undergo physical bench testing at an ATL. Tests must validate performance against strict mathematical and engineering tolerances:

3.1 Shaft Mechanical Tolerances & Drive Splines

- **Geometric Fit:** Male/female drive spline dimensions must match MPM-SPEC-101 profile within a tolerance of $\pm 0.012\text{ mm}$.
- **Torque Rating Verification:**
 - Baseline (77 kW at $1,800\text{ RPM}$): Continuous duty output $\geq 408.5\text{ N}\cdot\text{m}$ without tooth deflection $> 0.005\text{ mm}$.
 - Upscaled (1 MW at $1,800\text{ RPM}$): Continuous duty output $\geq 5,305\text{ N}\cdot\text{m}$.

3.2 Dynamic Balance & Harmonic Isolation

- **Rotational Unbalance:** Individual MPM shafts must achieve $G2.5$ balance grade per ISO 1940-1 at max operating speed ($3,600\text{ RPM}$).
- **Lateral RMS Velocity:** Maximum permissible housing vibration velocity $\leq 0.05\text{ mm/s}$ RMS under full rated load.
- **Torsional Vibration Transmission:** Aggregator gearboxes must reduce inter-engine peak torque oscillation to $\leq 2.0\%$ across the 5 Hz to 500 Hz harmonic spectrum.

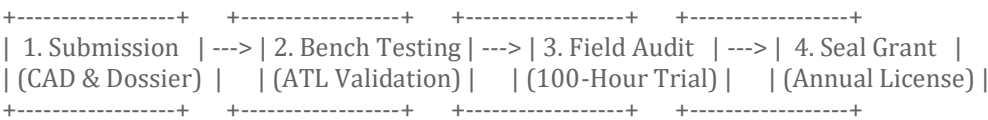
3.3 Thermal Dissipation & Fluid Interfaces

- **Continuous Duty Thermal Loop:** Pressurized lubrication and brine/seawater cooling systems must sustain 100% continuous rated duty (1 MW thermal rejection threshold for 10-MPM array) without exceedance of 85°C fluid operating temperature.
- **Quick-Disconnect Flanges:** Fluid fittings must demonstrate zero leakage under $1.5\times$ maximum operational pressure per ISO 16028.

3.4 Operational Telemetry & Carbon Offsets

- **Encryption & Data Integrity:** On-board telemetry processors must implement AES-256 encrypted hardware security modules (HSMs) for reporting mechanical work output (7.2 GWh/month equivalent baseline) per MPM-SPEC-403.

4. Certification Lifecycle & Audit Workflow

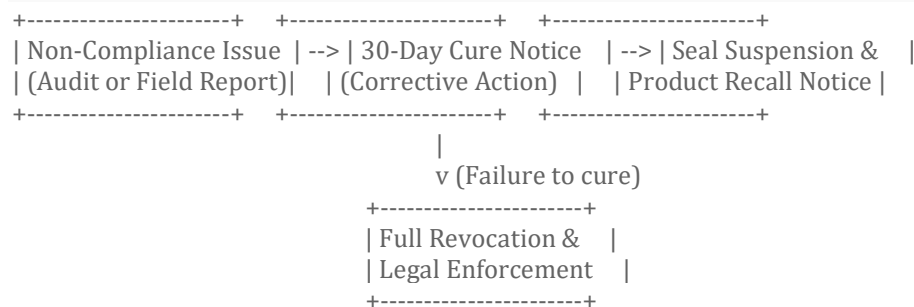


1. **Phase 1: Dossier & Geometric Review:** Manufacturer submits mechanical CAD models, material certifications, and finite element analysis (FEA) simulations to the CCB.
2. **Phase 2: Laboratory Bench Testing:** Candidate hardware is installed on an ATL test bench. Physical units undergo 50 hours of continuous full-load testing, shock load cycles, and thermal heat soak.
3. **Phase 3: Field Trial & Harmonic Verification:** Candidate adapters or modules execute a 100-hour continuous field demonstration under multi-engine load variations to verify resonance isolation.

4. **Phase 4: Seal Grant & Registry Listing:** Upon meeting all metric criteria, the CCB issues a Certificate of Compliance and lists the product in the official Consortium Public Registry.

5. Non-Compliance, Audit Enforcement & Sanctions

To prevent substandard hardware from damaging aggregator gearboxes or undermining market trust, the CCB maintains rigorous enforcement protocols:



5.1 Audit Triggers

- **Routine Surveillance:** Annual mandatory field audits of 10% of all certified production lines.
- **Field Incident Reports:** Mandatory investigation triggered if two or more independent field operators report destructive dynamic harmonics or premature spline wear.

5.2 Sanction Levels

1. **Warning & Investigation Notice:** Issued upon preliminary finding of minor variance (e.g., non-critical telemetry schema formatting error).
2. **30-Day Cure Notice:** Issued for physical non-compliance (e.g., vibration velocity exceeding 0.05 mm/s RMS or fluid fitting seepage).
3. **Seal Suspension:** Immediate suspension of certification seal and public notice if non-compliance poses structural or operational safety risks.
4. **Permanent Revocation & Disqualification:** Issued by a 2/3 vote of the BoD for fraudulent submission, willful violation of FRAND/RF IP licensing terms, or failure to rectify non-compliance within the 30-day window.

6. Regulatory Appeals Procedure

1. **Filing:** A manufacturer may appeal a CCB suspension or revocation notice within 15 calendar days of issuance.
2. **Technical Review Panel:** An independent three-member panel from the Technical Steering Committee (TSC), excluding members with financial conflicts of interest, conducts a technical re-audit.
3. **Final Determination:** The BoD reviews the panel's findings and issues a binding final determination via simple majority vote (or 2/3\$ supermajority to override panel recommendations).